EE2001 Electrical Circuits I:

Welcome and Introduction to the Course

02 October 2023, Monday

Asst. Prof. Dr. M. Alparslan Zehir



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Brief Personal Info:



Education:

BSc, MSc and PhD: Istanbul Technical University (ITU), Dpt. of Electrical Engineering

Research and Teaching Experience:

2011-2019: ITU Electrical Engineering, Research & Teaching Assistant

Postdoctoral Researcher Experience:

2019-2020: University College Cork, MaREI Centre (Ireland)

Areas of Interest:

- **Power Systems**
- **Smart Grids**
- **Microgrids**
- **Virtual Power Plants**
- **Renewable Energy**
- **Energy Storage**
- **Electric Vehicle Charging**
- **Electricity Markets**
- **Energy Management**
- **Building Automation Systems**

EE2001 Course Information:

Class Schedule:

Theoretical Lessons (every week):

Monday 09.30-11.30 – M1 building Z06 (2 hours, lesson) (only **physical**)

Tuesday 09.30-10.30 – M1 building Z01 (1 hour, lesson) (only **physical**)

Laboratory Experiments (only on certain weeks that will be announced):

Thursday Groups:

Thursday 15.00-16.50 (2 hours, laboratory & practice by the Research Assistant)

Friday Groups:

Friday 14.00-15.50 (2 hours, laboratory & practice by the Research Assistant)

Course Assistant: Res. Asst. Oguzhan Ciftci

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EE2001 Course Information:

Grades:

Laboratory Reports	: 20%
Midterm Exam	: 40%
Final Exam	: 40%

EE2001 Course Information:

There is no bonus grade for attending to the lessons.

Mandatory minimum 80% attendance to **physical laboratory experiments and lab report submission** (minimum 4 of the 5 experiments)

Mandatory minimum 70% attendance to **theoretical courses**

First time takers: Fail with DZ grade if the mandatory attendance criterions are not met.

People who **took the course before and failed with DZ:** Can fail again with DZ **if the mandatory attendance criterions are not met.**

People who **took the course before and failed with FF or took higher grades**: They do not fail with DZ even if the mandatory attendance criterions are not met. **HOWEVER** the previous years' lab grades or other grades **are NOT counted**. Therefore, if they do not attend the experiments this year, **20% of their grade will be missing** which may lead to FF or lower grades then their expectations.

EE2001 Course Information:

Text Book:

James W. Nilsson, Susan A. Riedel: "**Electric Circuits**", 9th Ed. (**or other editions**), Prentice Hall, New Jersey, 2011.

EE2001 Course Information:

Course Outline:

- Circuit elements and Kirchhoff's laws,
- Analysis of resistive circuits,
- Network theorems,
- Capacitors and inductors,
- Analysis of first and second order circuits,
- Operational Amplifiers,
- Complex variables.
- Sinusoidal steady-state analysis.
- Laboratory work

This course **have to be passed** to be able to take **EE2002 Electrical Circuits II and EE3011 Electronics I.**

MOREOVER, **EE2002 have to be passed** to be able to take **EE3042 Energy Conversion**

EE3011 have to be passed to be able to take **EE3012 Electronics II**

EE2001 Course Information:

Course Objectives:

- Have the ability to use the techniques in electrical circuit analysis

- Learn how to comment on electrical circuits by the help of computer aided design tools

- Learn the basic principles of electrical circuit design

- Make electrical measurements and calculations by the help of experiments

- Learn to write a lab report and how to make comment on the results, individually and as a group

EE2001 Course Information:

Course Outcomes:

- Relate fundamental concepts in electrical engineering,

- Distinguish linear and nonlinear circuit modeling,

- Identify basic circuit components,

- Apply Kirchhoff's laws,

- Analyze linear DC and AC networks using fundamental circuit theorems.

EE2001 Course Information:

Tentative Schedule:

Week	Date	Subjects	The Related Chapter from the Book
1	Oct. 2, 3	Introduction and Circuit Variables	1
2	Oct. 9, 10	Circuit Elements	2
3	Oct. 16, 17	Simple Resistive Circuits	3
4	Oct. 23, 24	Node Voltage Analysis	4
5	Oct. 30, 31	Mesh Current Analysis	4
6	Nov. 6, 7	Thevenin and Norton Equivalents	4
7	Nov. 13, 14	The Operational Amplifies	5
	Midterm Week		
8	Nov. 27, 28	Inductance and Capacitance	6
9	Dec. 4, 5	Response of First-Order RL and RC Circuits	7
10	Dec. 11, 12	Natural and Step Responses of RLC Circuits	8
11	Dec. 18, 19	Sinusoidal Steady-State Analysis	9
12	Dec. 25, 26	Continuation to: Sinusoidal Steady-State Analysis	9
13	Jan. 2	Sinusoidal Steady-State Power Calculations	10
14	Jan. 8, 9	Continuation to: Sinusoidal Steady-State Power Calculations	10

EE2001 Electrical Circuits I:

Circuit Variables

02 October 2023, Monday

Outline



1. **Five Major Classification of Electrical Systems**
2. **Circuit Theory**

Announcement

- **There is no laboratory lesson this week.**

1 Five Major Classification of Electrical Systems (they have **electric circuits** in common)

- **Communication systems** (use electrical signals to generate, transmit, distribute information)
  - TV equipment (cameras, transmitters, receivers), radio telescopes, satellite systems, radar systems, telephone systems (microphone, telephone, switching center, microwave station, transmission antenna, communication satellite).
- **Computer systems** (use electrical signals to process information ranging from word processing to mathematical equations)
  - Networks of microcircuits or integrated circuits – postage-stampsized assemblies of hundreds, thousands or millions of electrical components, operating at high speed and power levels close to physical limits
- **Control systems** (use electrical signals to regulate processes)
  - Control of temperatures-pressures and flow rates of an oil rafinery, motors-doors-lights in elevators, fuel-air mixture in a fuel-injected automobile engine, autopilot and autolanding systems

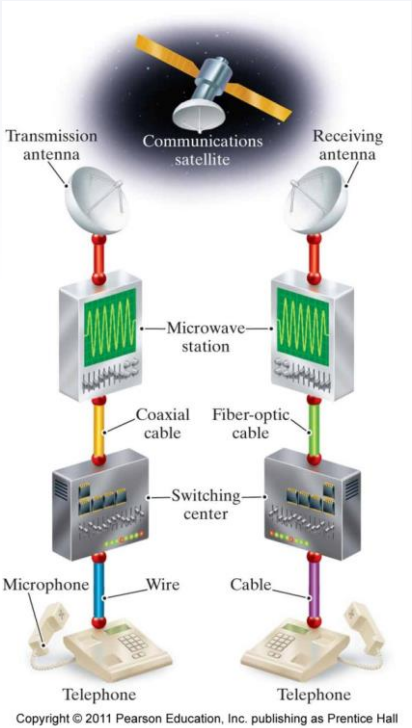
1 Five Major Classification of Electrical Systems (they have **electric circuits** in common)

- **Power systems** (generate, transmit and distribute electric power)
  - Power plants (thermal -coal, natural gas, oil-, hydropower, nuclear, wind, solar), interconnected national and internationally interconnected grids of conductors and other equipment, stable operation, energy quality, resiliency and many other aspects as aims in design and operation
- **Signal-processing systems** (act on electrical signals that represent information)
  - Transform signals and the information contained in them into a more suitable form (image-processing of massive quantities of data from orbiting weather satellites to reduce the amount of data to a manageable level and transform the remaining data into video image for further use or computerized tomography scan for diagnosis of disease and injury)

High interaction among the engineering disciplines related to these systems: Communication engineers use computers to control flow of information, computer contain control systems, power systems require extensive communication systems to operate safely and reliably, a signal-processing system usually contains a communications link, a computer and a control system.

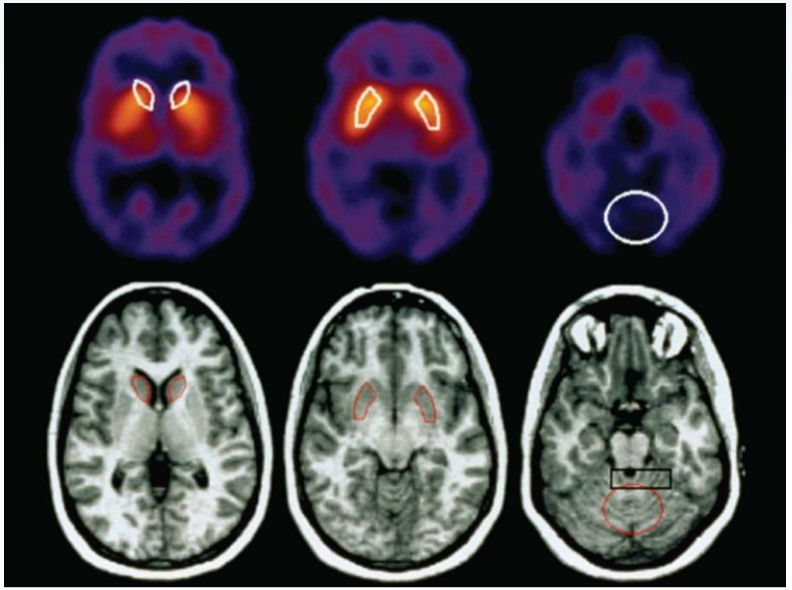
Five Major Classification of Electrical Systems (they have **electric circuits** in common)

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A telephone system


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A computerized tomography of an adult head

2 Circuit Theory

- An **electric circuit** is a mathematical model that **approximates** the behavior of an actual electrical system. It is commonly used to refer to an actual electrical system as well as the model that represents it.
- The **models**, the **mathematical techniques** and the **language of circuit theory** form the intellectual framework of your future engineering activities.
- Circuit theory is **a special case of electromagnetic field theory**: the study of static and moving electric charges, under the assumptions of:
  - Electrical effects **happen instantaneously** (there is no delay, at or near speed of light),
  - **The net charge** on every content in the system is zero,
  - There is **no magnetic coupling between** the components in a system (but it can occur within a component).

2 Circuit Theory

- Using circuit theory provides **simple solutions (of sufficient accuracy)** to problems that would become **much more complicated** if we were to use electromagnetic field theory.

The International System of Units (SI)

3

- Engineers **compare** theoretical results to experimental results and compare competing engineering designs **using quantitative measures**.
- Modern engineering is a **multidisciplinary profession,** in which teams of engineers work together on projects and they can **communicate their results** in a meaningful way if they all use **the same units of measure**.
- **The international system of units (SI)** is a unifying system of units used by many societies and engineers throughout the world.
- The SI units are based on **seven defined base quantities**. One of them is **electric current**.
- The other quantities are **derived** from the **base quantities**.
- In case of **too small or too large quantities**, **standard prefixes** corresponding to **powers of 10** are applied to the basic unit. Engineers often use only the **prefixes for powers divisible by 3** (mili (10^{-3}), micro (10^{-6}), nano (10^{-9}), pico (10^{-12}) and
- kilo (10^{3}), mega (10^{6}), giga (10^{9}), tera (10^{12}) are mostly used)

The International System of Units (SI)

3

TABLE 1.1 The International System of Units (SI)

Quantity	Basic Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	degree kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

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End of the Lesson (02.10.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist

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